

Problem 6

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Problem 1 Find the error in the following “solution”, correct the error (or errors) and evaluate the integral.

Find $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x) \sqrt{1 - \cos^2(x)} dx$.

”Solution”:

Since $\sin(x) = \sqrt{1 - \cos^2(x)}$, let $u = \sin(x)$. Then $du = \cos(x)dx$;

when $x = -\frac{\pi}{2}$, $u = \sin\left(-\frac{\pi}{2}\right) = -1$; when $x = \frac{\pi}{2}$, $u = \sin\left(\frac{\pi}{2}\right) = 1$. Therefore

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(x) \sqrt{1 - \cos^2(x)} dx = \int_{-1}^1 u \cdot du = 0,$$

due to symmetry (integrating an odd function over a symmetric interval, $[-1, 1]$).